

METAR: Conducting Meta-Analysis with R

Full Program, September-November 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Conducting Meta-Analysis with R

Format: Online, cohort-based

Duration: 3 weeks, 9 sessions, 2 hours per session, 18 contact hours

Dates: 09/11/2026 – 27/11/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00)

Time: Morocco time

Price: 200€ (2000 MAD)

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

METAR is a practical and methodologically rigorous introduction to conducting meta-analysis with R. It covers the complete evidence-synthesis workflow: formulating a review question, building a search strategy, developing an analysis-ready extraction table, computing effect sizes, fitting fixed-effect and random-effects models, assessing heterogeneity, producing forest plots, exploring moderators, assessing publication bias, rating certainty of evidence, and preparing a reproducible Quarto report.

The course emphasizes transparent reporting, correct interpretation of pooled estimates, reproducible analysis, and publication-quality tables and figures.

1.2 Target Audience

Clinicians, researchers, PhD students, public health professionals, and evidence synthesis analysts. Basic epidemiology or clinical research experience recommended.

1.3 Prerequisites

Basic familiarity with R (as covered in BIOSTATR) is recommended but not mandatory.

1.4 Learning Objectives

By the end of METAR, participants will be able to:

1. formulate a meta-analysis question using the PICO framework and write a review protocol;
2. build and document a reproducible search and study-selection workflow;
3. structure extracted study data and risk-of-bias assessments for analysis in R;
4. compute effect sizes for binary and continuous outcomes;
5. fit fixed-effect and random-effects meta-analysis models with prediction intervals;
6. quantify and explore heterogeneity using Q , I^2 , and τ^2 ;
7. produce publication-quality forest plots and interpret pooled estimates;
8. conduct subgroup analysis and meta-regression, and assess publication bias;
9. rate certainty of evidence with GRADE and produce a reproducible meta-analysis report with Quarto.

2 Detailed Schedule

Table 1: METAR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	09/11/2026 (Mon)	18h00- 20h00	Systematic reviews and the review protocol	Evidence hierarchy, PICO, PRISMA workflow, review protocol, eligibility criteria
2	1	11/11/2026 (Wed)	18h00- 20h00	Search strategy and study selection	Databases, search strings, deduplication, screening, PRISMA flow diagram
3	1	13/11/2026 (Fri)	18h00- 20h00	Data extraction and risk of bias	Extraction tables, risk-of-bias tools, robvis , importing and cleaning data in R, unit harmonization, consistency checks
4	2	16/11/2026 (Mon)	18h00- 20h00	Effect sizes for binary and continuous outcomes	Odds ratio, risk ratio, risk difference, mean difference, standardized mean difference, Hedges' g
5	2	18/11/2026 (Wed)	18h00- 20h00	Fixed-effect and random-effects models	Inverse-variance weighting, common-effect model, DerSimonian-Laird, REML, Paule-Mandel, prediction intervals
6	2	20/11/2026 (Fri)	18h00- 20h00	Heterogeneity and sensitivity analysis	Cochran's Q , I^2 , τ^2 , heterogeneity CIs, Baujat plots, leave-one-out analysis, influence diagnostics
7	3	23/11/2026 (Mon)	18h00- 20h00	Forest plots and result interpretation	Forest plot anatomy, study weights, summary diamond, prediction interval, subgroup forest plots
8	3	25/11/2026 (Wed)	18h00- 20h00	Subgroup analysis, meta-regression, and publication bias	Meta-regression, bubble plots, funnel plots, Egger/Begg/Peters tests, trim-and-fill

No.	Week	Date	Time	Topic	Main Content
9	3	27/11/2026 (Fri)	18h00- 20h00	Certainty of evidence and reproducible reporting	GRADE, Quarto report, PRISMA diagram, risk-of-bias plot, effect-size table, forest plot, final project

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

The course ends with a short reproducible Quarto report or mini-project, presented and discussed in the final session.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required.

Attendance: Full attendance at all 9 sessions is recommended given the cumulative structure.